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Center of Mass & Change of Variables

Suppose objects of masses m_1, m_2,...,m_n be arranged along a line at coordinates x_1, x_2,...,x_n respectively and the balance point or center of mass is at $\overline{x}$. Each mass contributes a "torque" of $m_i g(x_i - \overline{x})$ (where g is the magnitude of acceleration due to gravity) and to balance,

$$\sum_{i=1}^{n} m_i g(x_i - \overline{x}) = 0 \rightarrow \overline{x} = \frac{m_1 x_1 + m_2 x_2 + \cdots + m_n x_n}{m_1 + m_2 + \cdots + m_n}.$$

Now consider a continuous (linear) density function $\rho(x)$ for an object occupying the interval $[a, b]$. One can take n subintervals of the "rod," each of width Δx, and use sample points $x_i = a + i\Delta x$, to get a limit for the balance condition for finding the center of mass $\overline{x}$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \rho(x_i)\Delta x g(x_i - \overline{x}) = 0 \rightarrow$$

$$\lim_{n\to\infty} \sum_{i=1}^{n} \rho(x_i)\Delta x (x_i - \overline{x}) = 0 \rightarrow$$

$$\overline{x} = \lim_{n\to\infty} \frac{\sum_{i=1}^{n} \rho(x_i) x_i \Delta x}{\sum_{i=1}^{n} \rho(x_i)\Delta x}.$$

Thus the balance condition becomes,

$$\overline{x} = \frac{\int_a^b x\rho(x)dx}{\int_a^b \rho(x)dx}.$$

Note that the denominator is the mass! The numerator is called the moment.

Now let a "plate" $R = \{(x,y)|a \leq x \leq b, g(x) \leq y \leq f(x)\}$ have a continuous (planar) density function $\rho(x,y)$.

To compute the center of mass, $(\overline{x}, \overline{y})$, let's consider how to compute $\overline{x}$ (it is analogous for $\overline{y}$):

Suppose we partition $[a,b]$ into n subintervals of equal width Δx and let $x_i = a + i\Delta x$. We consider the torque about the y-axis of the vertical part of the plate that lies along the line where x is a constant as though it was a point mass at $(x,0)$. Then $\lim_{n\to\infty} \sum_{i=1}^{n} x_i \left(\int_{g(x_i)}^{f(x_i)} \rho(x_i, y) dy \right) \Delta x$ is the moment about the y-axis. So

$$M_y = \int \int_R x\rho(x,y) dA.$$

Similarly, the moment about the x-axis is

$$M_x = \int \int_R y\rho(x,y) dA.$$

Note: These formulas extend to all regions, not just ones that can be described with $a \leq x \leq b$ and $g(x) \leq y \leq f(x)$.

Thus:

$$\overline{x} = \frac{M_y}{m} \text{ and } \overline{y} = \frac{M_x}{m} \text{ where the mass is } m.$$

A plate occupies the region R in the first quadrant where $x^2 + y^2 \leq 4$ (x, y are in cm) with the density function $\rho(x,y) = 5 - xy$ grams per cubic cm. Let's find the mass and the center of mass.

Due to the reflective symmetry of region over the line $y = x$, and since x, y play symmetric roles in $\rho(x,y)$, we know that $\overline{x} = \overline{y}$. The mass is

$$\begin{aligned} m &= \int_0^{\frac{\pi}{2}} \int_0^2 (5 - r^2 \sin(\theta)\cos(\theta))\, r\, dr\, d\theta \\ &= \frac{\pi}{2} \left(\frac{5}{2} r^2 \Big|_0^2 \right) - \left(\frac{1}{2} \sin^2(\theta) \Big|_0^{\frac{\pi}{2}} \right) \left(\frac{1}{4} r^4 \Big|_0^2 \right) \\ &= 5\pi - 2. \end{aligned}$$

The moment about the y-axis is

$$\begin{aligned} M_y &= \int_0^{\frac{\pi}{2}} \int_0^2 r\cos(\theta)\left(5 - r^2\sin(\theta)\cos(\theta)\right) r\, dr\, d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^2 5r^2\cos(\theta) - r^4\sin(\theta)\cos^2(\theta)\, dr\, d\theta \\ &= \left(\sin(\theta)|_0^{\frac{\pi}{2}}\right)\left(\frac{5}{3}r^3\Big|_0^2\right) - \left(-\frac{1}{3}\cos^3(\theta)\Big|_0^{\frac{\pi}{2}}\right)\left(\frac{1}{5}r^5\Big|_0^2\right) \\ &= \left(\frac{40}{3}\right) - \left(\frac{1}{3}\right)\left(\frac{32}{5}\right) \\ &= \frac{168}{15} = \frac{56}{5}. \end{aligned}$$

So the center of mass is $\left(\frac{56}{5(5\pi-2)}, \frac{56}{5(5\pi-2)}\right) \approx (0.8, 0.8)$.

Suppose a solid occupies a region E in xyz-sapce with a continuous density function $\rho(x, y, z)$ on E. Then the moments are about planes. The center of mass $(\overline{x}, \overline{y}, \overline{z})$ satisfies:

$$\overline{x} = \frac{M_{yz}}{m},$$

$$\overline{y} = \frac{M_{xz}}{m},$$

$$\overline{z} = \frac{M_{xy}}{m},$$

where $m = \int\int\int_E \rho(x, y, z)dV$ is the mass and

$$M_{yz} = \int\int\int_E x\rho(x, y, z)dV,$$

$$M_{xz} = \int\int\int_E y\rho(x, y, z)dV,$$

$$M_{xy} = \int\int\int_E z\rho(x, y, z)dV.$$

A change of variables in a double integral is a transformation that relates two sets of variables, (u, v) and (x, y), where the region R in the xy-plane corresponds to a region S in the uv-plane.

Suppose $x = g(u, v)$ and $y = h(u, v)$ defines transformation where g, h have continuous first order partial derivatives over S and are one-to-one on the interior of S. The Jacobian determinant is computed in the following way:

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}.$$

Change of Variables Theorem (2D): Suppose $f(x, y)$ is continuous on a region R in the xy-plane that transforms (with the assumptions above) into a region S in the uv-plane. Then

$$\int\int_R f(x,y)dA = \int\int_S f(g(u,v), h(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dA.$$

(We skip the proof.)

Let's evaluate $\int\int_R y^2 dA$ where R is the region in the first quadrant bounded by $xy = 1$, $xy = 4$, $\frac{y}{x} = 1$ and $\frac{y}{x} = 3$.

We change the variables with $u = xy$ and $v = \frac{y}{x}$ because in the uv-plane the transformed region is $[1, 4] \times [1, 3]$.

We have to solve for x, y in terms of u, v. Since $u/v = x^2$ and $uv = y^2$ and x, y, u, v are all positive, $x = \sqrt{\frac{u}{v}}$ and $y = \sqrt{uv}$. So

$$\frac{\partial(x,y)}{\partial(u,v)} = det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = det \begin{pmatrix} \frac{1}{2\sqrt{uv}} & -\frac{\sqrt{u}}{2\sqrt{v^3}} \\ \frac{\sqrt{v}}{2\sqrt{u}} & \frac{\sqrt{u}}{2\sqrt{v}} \end{pmatrix} = \frac{1}{4v} + \frac{1}{4v} = \frac{1}{2v}.$$

So $\int\int_R y^2 dA = \int_1^3 \int_1^4 uv\frac{1}{2v}\, du\, dv = \int_1^4 u\, du = \frac{15}{2}$.

Suppose $x = g(u, v, w)$, $y = h(u, v, w)$ and $z = \ell(u, v, w)$ is a transformation of a solid B in uvw-space into a solid E in xyz-space that has continuous 1st order partials and is 1-1 on the interior. The Jacobian determinant is computed in the following way:

$$J(u, v, w) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}.$$

Change of Varaibles Theorem (3D) Suppose $f(x, y, z)$ is continuous on a solid region E in xyz-space that transforms (with the assumptions above) into a solid region B in uvw-space. Then

$$\int\int\int_E f(x, y, z)dV = \int\int\int_B f(g(u, v, w), h(u, v, w), \ell(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| dV.$$